STRICT NORMAL CROSSINGS DIVISORS CHARACTERIZED BY REDUCEDNESS AND REGULAR INTERSECTIONS OF COMPONENTS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original definition 0BI9: sources/stacks-chow/tags.txt, source lines 14482–14482. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NQ: sources/stacks-chow/tags.txt, source lines 868–868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NU: sources/stacks-chow/tags.txt, source lines 872–872. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 033M: sources/stacks-chow/tags.txt, source lines 4061–4061. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NA: sources/stacks-chow/tags.txt, source lines 852–852. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $D \subset X$ be an effective Cartier divisor. Let $D_i \subset D$, $i \in I$ be its irreducible components viewed as reduced closed subschemes of X. The following are equivalent*

1. *D is a strict normal crossings divisor, and*
2. *D is reduced, each D_i is an effective Cartier divisor, and for $J \subset I$ finite the scheme theoretic intersection $D_J = \bigcap_{j \in J} D_j$ is a regular scheme each of whose irreducible components has codimension $|J|$ in X.*

Proof. Assume D is a strict normal crossings divisor. Pick $p \in D$ and choose a regular system of parameters $x_1, \ldots, x_d \in \mathfrak{m}_p$ and $1 \leq r \leq d$ as in Definition 0BI9. Since $\mathcal{O}_{X,p}/(x_i)$ is a regular local ring (and in particular a domain) we see that the irreducible components $D_1, \ldots, D_r$ of D passing through p correspond 1-to-1 to the height one primes $(x_1), \ldots, (x_r)$ of $\mathcal{O}_{X,p}$. By Algebra, Lemma 00NQ we find that the intersections $D_{i_1} \cap \ldots \cap D_{i_s}$ have codimension s in an open neighbourhood of p and that this intersection has a regular local ring at p. Since this holds for all $p \in D$ we conclude that (2) holds.

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Assume (2). Let $p \in D$. Since $\mathcal{O}_{X,p}$ is finite dimensional we see that p can be contained in at most $\dim(\mathcal{O}_{X,p})$ of the components D_i. Say $p \in D_1, \ldots, D_r$ for some $r \geq 1$. Let $x_1, \ldots, x_r \in \mathfrak{m}_p$ be local equations for $D_1, \ldots, D_r$. Then x_1 is a nonzerodivisor in $\mathcal{O}_{X,p}$ and $\mathcal{O}_{X,p}/(x_1) = \mathcal{O}_{D_1,p}$ is regular. Hence $\mathcal{O}_{X,p}$ is regular, see Algebra, Lemma 00NU. Since $D_1 \cap \ldots \cap D_r$ is a regular (hence normal) scheme it is a disjoint union of its irreducible components (Properties, Lemma 033M). Let $Z \subset D_1 \cap \ldots \cap D_r$ be the irreducible component containing p. Then $\mathcal{O}_{Z,p} = \mathcal{O}_{X,p}/(x_1, \ldots, x_r)$ is regular of codimension r (note that since we already know that $\mathcal{O}_{X,p}$ is regular and hence Cohen-Macaulay, there is no ambiguity about codimension as the ring is catenary, see Algebra, Lemmas 00NQ and 00NA). Hence $\dim(\mathcal{O}_{Z,p}) = \dim(\mathcal{O}_{X,p}) - r$. Choose additional $x_{r+1}, \ldots, x_n \in \mathfrak{m}_p$ which map to a minimal system of generators of $\mathfrak{m}_{Z,p}$. Then $\mathfrak{m}_p = (x_1, \ldots, x_n)$ by Nakayama's lemma and we see that D is a normal crossings divisor. $\square$